

# WEEK-1(OS)

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    // 1. Accept a Linux command as input from the user
    printf("Enter a Linux command to execute: ");
    if (fgets(command, sizeof(command), stdin) == NULL) {
        perror("Error reading input");
        return 1;
    }

    // Remove the trailing newline character from fgets
    command[strcspn(command, "\n")] = '\0';

    // 2. Create a child process using fork()
    pid_t pid = fork();

    if (pid < 0) {
        // Fork failed
        perror("Fork failed");
        return 1;
    }
    else if (pid == 0) {
        // --- CHILD PROCESS ---
        // 5. Display the Process ID (PID) of the child process
        printf("\n[Child] Child Process PID: %d\n", getpid());
        printf("[Child] Executing command: %s\n\n", command);

        // 3. Execute the command using an appropriate exec() system call
        // execlp searches the PATH environment variable for the command
        execlp(command, command, (char *)NULL);

        // If execlp returns, an error occurred
        perror("Execution failed");
        exit(1);
    }
    else {
        // --- PARENT PROCESS ---
        // 5. Display the Process ID (PID) of the parent process
        printf("[Parent] Parent Process PID: %d\n", getpid());

        printf("[Parent] Parent Process PID: %d\n", getpid());
        printf("[Parent] Waiting for the child process to complete...\n");

        // 4. Allow the parent process to wait for the child using wait()
        int status;
        wait(&status);

        printf("\n[Parent] Child process finished execution.\n");
    }

    return 0;
}
```

# Output

```
ashwika@ashwikapadamata:~$ vi week1.c
ashwika@ashwikapadamata:~$ gcc week1.c -o week1
ashwika@ashwikapadamata:~$ ./week1
Enter a Linux command to execute: ls
[Parent] Parent Process PID: 5477
[Parent] Waiting for the child process to complete...

[Child] Child Process PID: 5478
[Child] Executing command: ls
add          arithmetic.c : childprocess.c  eliminatezombie.c  file          mul          process1  week1      zombie.c
addition.c   child          dir1          exe             file.c        nano.2546.save process1.c week       weekthree.c
arithmetic   child.c        dir1          exec.c          fl.txt.save   parent       switch    week.c     weekthree.c
arithmetic.c childprocess   eliminatezombie fl.txt          fl.txt.save   parent.c     switch.c  week1      zombie

[Parent] Child process finished execution.
```